

ELITE IGCSE ACADEMY

Student Past Paper Solutions

Jun 2022 P1HR Worked Solutions

Jun 2022 | P1HR

31 questions ■ question image followed by worked solution

PREPARED BY

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STUDENT EDITION

Worked answers included

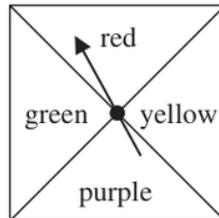
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PAST PAPER QUESTION**Q#: 1****Marks: 4****TOPIC: Probability Toolkit**

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- 1** Here is a biased spinner.



When the spinner is spun once, the probabilities that it lands on red or on yellow or on green are given in the table.

Colour	red	yellow	purple	green
Probability	0.25	0.2		0.2

- (a) Work out the probability that the spinner lands on red or on yellow.

.....
(1)

Yang is going to spin the spinner 300 times.

- (b) Work out an estimate for the number of times the spinner will land on purple.

.....
(3)

(Total for Question 1 is 4 marks)

PAST PAPER SOLUTION

Q#: 1

Marks: 4

TOPIC: **Probability Toolkit**

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 2****Marks: 4****TOPIC: Sequences**

Jun 2022 P1HR - Jun 2022 | P1HR

- 2 In a warehouse there are two types of shelves, type **R** and type **S**.

These two types of shelves are arranged into shelving units that form a sequence of patterns.

Here are the first three terms in the sequence.

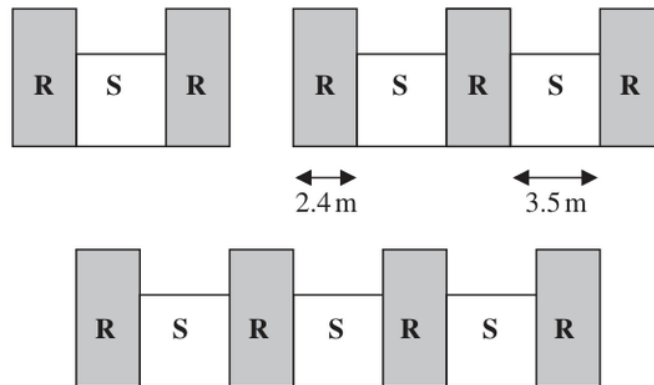


Diagram **NOT**
accurately drawn

The width of each type **R** shelf is 2.4 m and the width of each type **S** shelf is 3.5 m

- (a) Work out the total width of a shelving unit that has 6 type **R** shelves.

..... m
(2)

A shelving unit has n type **R** shelves.
The total width of this shelving unit is W metres.

- (b) Find an expression for W in terms of n
Give your answer in its simplest form.

$W =$
(2)

(Total for Question 2 is 4 marks)

PAST PAPER SOLUTION

Q#: 2

Marks: 4

TOPIC: **Sequences**

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 3****Marks: 3**TOPIC: **Statistics Toolkit**

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3 Here are five cards.

Each card has a number written on it.

15

7

-2

23

 x

The mean of the five numbers is 12

Work out the value of x $x = \dots\dots\dots$ **(Total for Question 3 is 3 marks)**

PAST PAPER SOLUTION

Q#: 3

Marks: 3

TOPIC: **Statistics Toolkit**

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 4****Marks: 4****TOPIC: Percentages**

Jun 2022 P1HR - Jun 2022 | P1HR

- 4** The language department of a college has 180 students.
Each student studies exactly one of French, German, Italian or Spanish.

15 students study French.

45% of the students study German.

Express the percentage of students studying Italian or Spanish as a percentage of those studying French or German.

.....%

(Total for Question 4 is 4 marks)

EA

PAST PAPER SOLUTION

Q#: 4

Marks: 4

TOPIC: **Percentages**

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 5****Marks: 5****TOPIC: Factorising**

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5 (a) Expand $3c^3(c + 4)$
(2)(b) (i) Factorise $x^2 + 8x - 9$
(2)(ii) Hence, solve $x^2 + 8x - 9 = 0$
(1)

(Total for Question 5 is 5 marks)

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PAST PAPER SOLUTION

Q#: 5

Marks: 5

TOPIC: **Factorising**

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 6****Marks: 3****TOPIC: Fractions**

Jun 2022 P1HR - Jun 2022 | P1HR

6 Show that $2\frac{2}{3} + 3\frac{3}{4} = 6\frac{5}{12}$

(Total for Question 6 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 6

Marks: 3

TOPIC: **Fractions**

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PAST PAPER QUESTION**Q#: 7****Marks: 3****TOPIC: Standard & Compound Units**

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- 7 The diagram shows a solid cylinder made from iron.

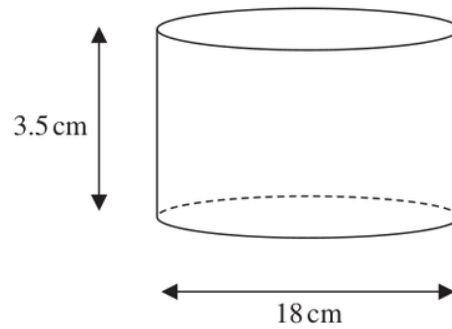


Diagram **NOT**
accurately drawn

The cylinder has diameter 18 cm and height 3.5 cm

The mass of the cylinder is 7.04 kg

Work out the density of the iron.

Give your answer in g/cm^3 correct to 2 significant figures.

..... g/cm^3

(Total for Question 7 is 3 marks)

PAST PAPER SOLUTION

Q#: 7

Marks: 3

TOPIC: **Standard & Compound Units**

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 8****Marks: 3****TOPIC: Compound Interest & Depreciation**

Jun 2022 P1HR - Jun 2022 | P1HR

- 8** Jane bought a new car for \$18 000
The car depreciates in value by 15% each year.
- Work out the value of the car at the end of 4 years.
Give your answer correct to the nearest \$

\$.....

(Total for Question 8 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 8

Marks: 3

TOPIC: **Compound Interest & Depreciation**

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 9****Marks: 2****TOPIC: Solving Inequalities**

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9 Solve the inequality $3 - 4x \leq 11$

(Total for Question 9 is 2 marks)

EA

PAST PAPER SOLUTION

Q#: 9

Marks: 2

TOPIC: **Solving Inequalities**

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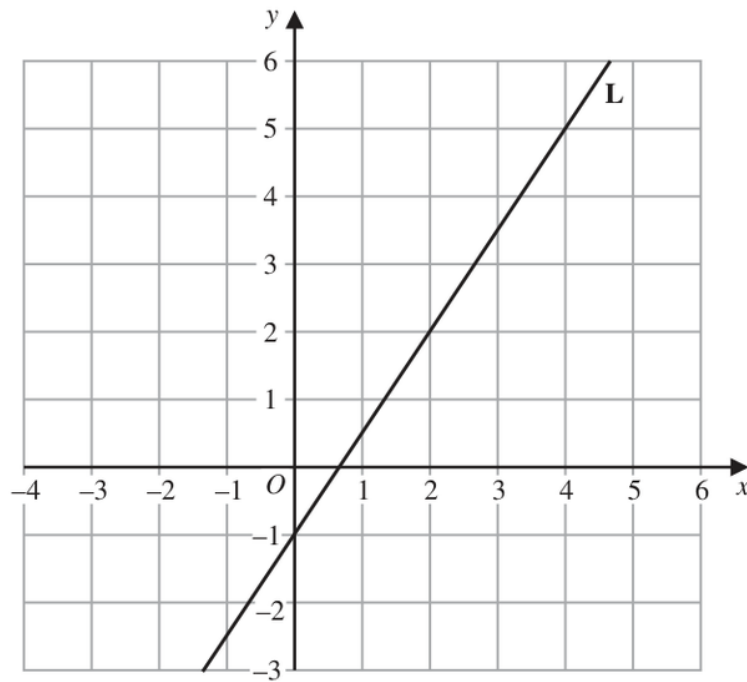
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EA

PAST PAPER QUESTION**Q#: 10****Marks: 3****TOPIC: Linear Graphs ($y = mx + c$)**

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10 Line **L** is drawn on the grid.Find an equation for **L**Give your answer in the form $y = mx + c$

(Total for Question 10 is 3 marks)

PAST PAPER SOLUTION

Q#: 10

Marks: 3

TOPIC: **Linear Graphs** ($y = mx + c$)

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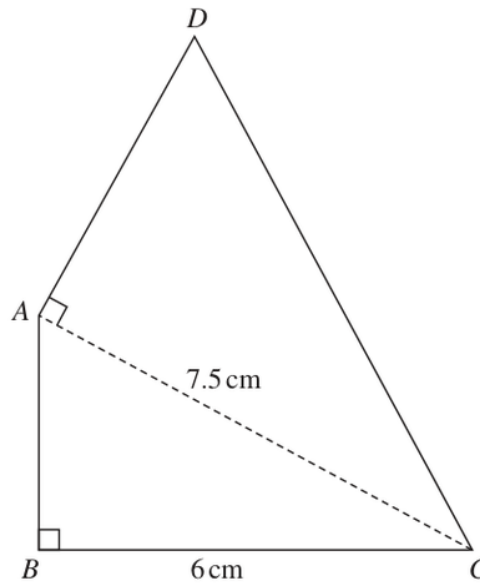
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EA

PAST PAPER QUESTION**Q#: 11****Marks: 6****TOPIC: Area & Perimeter**

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11 The diagram shows a quadrilateral $ABCD$ Diagram **NOT**
accurately drawnIn the diagram, ABC and DAC are right-angled triangles.

$$BC = 6 \text{ cm} \quad AC = 7.5 \text{ cm}$$

The area of quadrilateral $ABCD$ is 31.5 cm^2 Work out the length of AD

Question 11 continued.

..... cm

(Total for Question 11 is 6 marks)

PAST PAPER SOLUTION

Q#: 11

Marks: 6

TOPIC: **Area & Perimeter**

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 12****Marks: 3****TOPIC: Prime Factors, HCF & LCM**

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12 $P = 3^3 \times 5^2 \times 7$
 $Q = 3^2 \times 5 \times 7^2$

(a) Write down the highest common factor (HCF) of P and Q

.....
(1)

$P = 3^3 \times 5^2 \times 7$
 $Q = 3^2 \times 5 \times 7^2$

(b) Work out the value of $P^3 \times Q$

Give your answer in the form $3^x \times 5^y \times 7^z$ where x , y and z are positive integers.

.....
(2)

.....
(Total for Question 12 is 3 marks)

PAST PAPER SOLUTION

Q#: 12

Marks: 3

TOPIC: **Prime Factors, HCF & LCM**

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EA

PAST PAPER QUESTION**Q#: 13****Marks: 2****TOPIC: Statistics Toolkit**

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13 Here is the number of runs scored by a baseball team in each of its 15 games this season.

The number of runs have been arranged in order of size.

0 1 1 3 5 6 7 7 8 9 9 12 12 15 16

Work out the interquartile range of the number of runs.

(Total for Question 13 is 2 marks)

EA

PAST PAPER SOLUTION

Q#: 13

Marks: 2

TOPIC: **Statistics Toolkit**

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PAST PAPER QUESTION**Q#: 14****Marks: 4****TOPIC: Simultaneous Equations**

Jun 2022 P1HR - Jun 2022 | P1HR

14 Solve the simultaneous equations

$$3x - 5y = 25$$

$$4x + 3y = 14$$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

(Total for Question 14 is 4 marks)

EA

PAST PAPER SOLUTION

Q#: 14

Marks: 4

TOPIC: **Simultaneous Equations**

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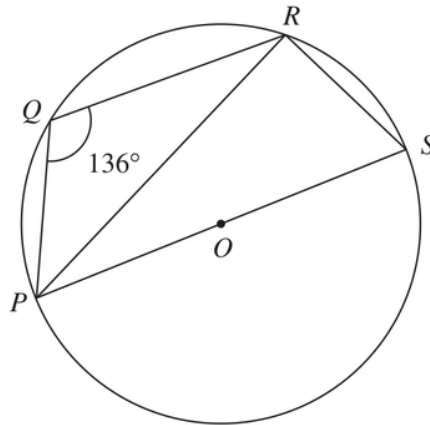
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EA

PAST PAPER QUESTION**Q#: 15****Marks: 3****TOPIC: Circle Theorems**

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15Diagram **NOT**
accurately drawn P , Q , R and S are points on a circle with centre O PS is a diameter of the circle.Angle $PQR = 136^\circ$ Work out the size of angle RPS

(Total for Question 15 is 3 marks)

PAST PAPER SOLUTION

Q#: 15

Marks: 3

TOPIC: **Circle Theorems**

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PAST PAPER QUESTION**Q#: 16****Marks: 6****TOPIC: Expanding Brackets**

Jun 2022 P1HR - Jun 2022 | P1HR

16 (a) Expand and simplify $(3x - 1)(x + 2)(3x + 1)$
(3)(b) Simplify fully $\left(\frac{2x^5}{8xy^2}\right)^{-2}$
(3)

(Total for Question 16 is 6 marks)

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Q#: 16

Marks: 6

TOPIC: **Expanding Brackets**

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PAST PAPER QUESTION**Q#: 17****Marks: 5****TOPIC: Sine, Cosine Rule & Area of Triangles**

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17 Here is a parallelogram $PQRS$, in which angle SPQ is acute.

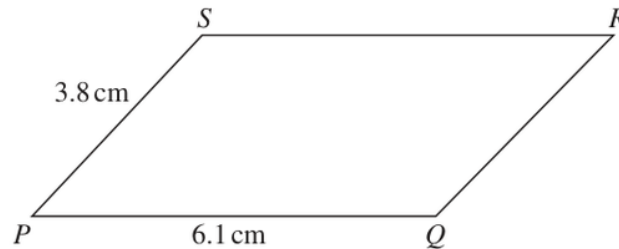


Diagram **NOT**
accurately drawn

$$PQ = 6.1 \text{ cm} \quad PS = 3.8 \text{ cm}$$

The area of the parallelogram is 18 cm^2

Work out the length of QS

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 17 is 5 marks)

PAST PAPER SOLUTION

Q#: 17

Marks: 5

TOPIC: **Sine, Cosine Rule & Area of Triangles**

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PAST PAPER QUESTION**Q#: 18****Marks: 4****TOPIC: 3D Pythagoras & Trigonometry**

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18 The diagram shows a cube $ABCDEFGH$ with sides of length 6 cm.

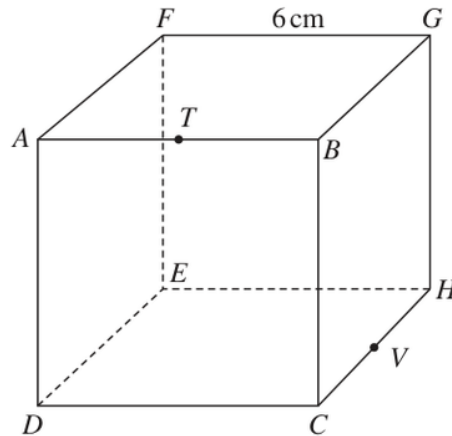


Diagram **NOT**
accurately drawn

T is the midpoint of AB and V is the midpoint of CH

Work out the distance from T to V in a straight line through the cube.

Give your answer in the form $\sqrt{a}$ cm where a is an integer.

..... cm

(Total for Question 18 is 4 marks)

PAST PAPER SOLUTION

Q#: 18

Marks: 4

TOPIC: **3D Pythagoras & Trigonometry**

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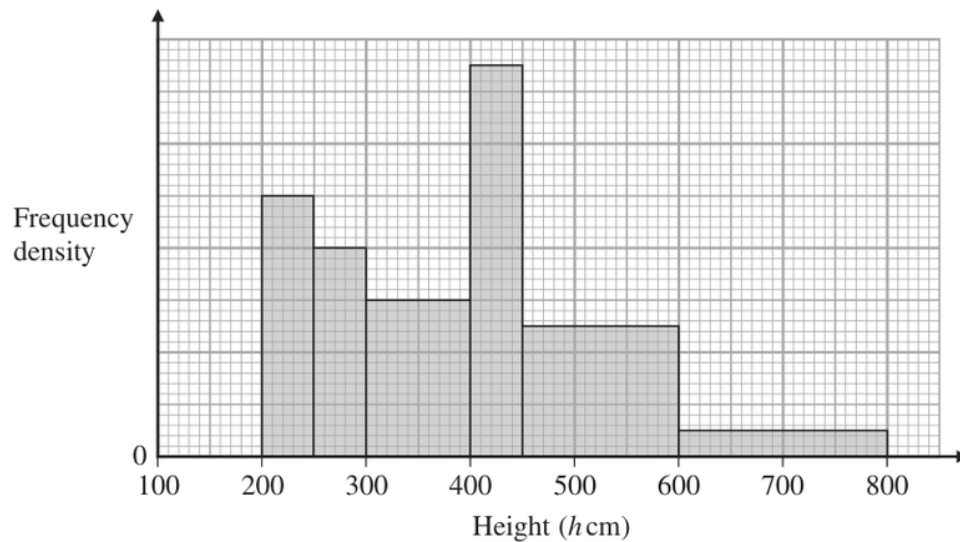
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EA

PAST PAPER QUESTION**Q#: 19****Marks: 3****TOPIC: Histograms**

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19 The histogram gives information about the height, h cm, of each tree in part of a forest.



There are no trees for which $h \leq 200$ and for which $h > 800$

The number of trees for which $300 < h \leq 400$ is 8 fewer than the number of trees for which $400 < h \leq 500$

Work out an estimate for the number of trees in this part of the forest that have a height greater than 500 cm.

(Total for Question 19 is 3 marks)

PAST PAPER SOLUTION

Q#: 19

Marks: 3

TOPIC: **Histograms**

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 20****Marks: 4****TOPIC: Area & Volume of Similar Shapes**

Jun 2022 P1HR - Jun 2022 | P1HR

20 The diagram shows two similar metal statues.

**A****B**

Diagram **NOT**
accurately drawn

The volume of statue **B** is 20% less than the volume of statue **A**

The surface area of statue **B** is $k\%$ less than the surface area of statue **A**

Work out the value of k

Give your answer correct to 3 significant figures.

$k = \dots\dots\dots$

(Total for Question 20 is 4 marks)

PAST PAPER SOLUTION

Q#: 20

Marks: 4

TOPIC: **Area & Volume of Similar Shapes**

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 20****Marks: 4****TOPIC: Area & Volume of Similar Shapes**

Jun 2022 P1HR - Jun 2022 | P1HR

20 The diagram shows two similar metal statues.

**A****B**

Diagram **NOT**
accurately drawn

The volume of statue **B** is 20% less than the volume of statue **A**

The surface area of statue **B** is $k\%$ less than the surface area of statue **A**

Work out the value of k

Give your answer correct to 3 significant figures.

$k = \dots\dots\dots$

(Total for Question 20 is 4 marks)

PAST PAPER SOLUTION

Q#: 20

Marks: 4

TOPIC: **Area & Volume of Similar Shapes**

Jun 2022 P1HR - Jun 2022 | P1HR

WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 21****Marks: 4**TOPIC: **Surds**

Jun 2022 P1HR - Jun 2022 | P1HR

21 Express $\frac{3 + \sqrt{8}}{(\sqrt{2} - 1)^2}$ in the form $p + \sqrt{q}$ where p and q are integers.

Show each stage of your working clearly.

(Total for Question 21 is 4 marks)

EA

PAST PAPER SOLUTION

Q#: 21

Marks: 4

TOPIC: **Surds**

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 21****Marks: 4**TOPIC: **Surds**

Jun 2022 P1HR - Jun 2022 | P1HR

21 Express $\frac{3 + \sqrt{8}}{(\sqrt{2} - 1)^2}$ in the form $p + \sqrt{q}$ where p and q are integers.

Show each stage of your working clearly.

(Total for Question 21 is 4 marks)

EA

PAST PAPER SOLUTION

Q#: 21

Marks: 4

TOPIC: **Surds**

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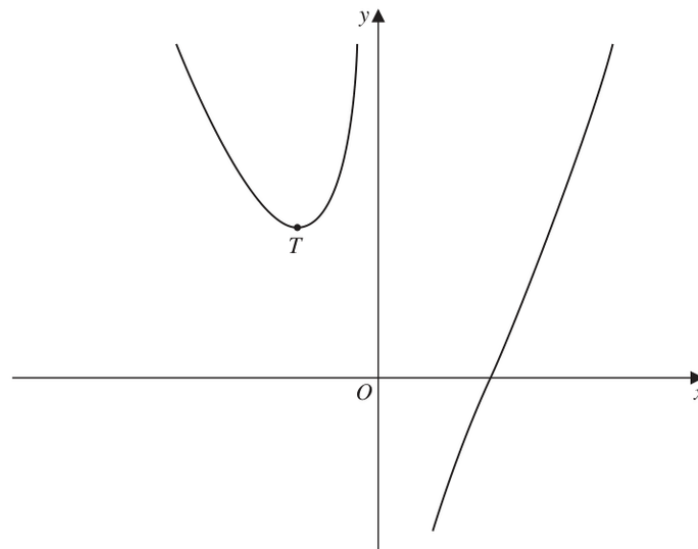
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EA

PAST PAPER QUESTION**Q#: 22****Marks: 7****TOPIC: Differentiation**

Jun 2022 P1HR - Jun 2022 | P1HR

- 22 The diagram shows a sketch of part of the curve with equation $y = x^2 - \frac{p}{x}$ where p is a positive constant.

Diagram **NOT**
accurately drawn

For all values of p , the curve has exactly one turning point and this turning point is a minimum shown as the point T in the sketch.

For the curve where the x coordinate of T is -3

- (a) find the value of p

$$p = \dots\dots\dots (4)$$

The line with equation $y = k$ is a tangent to the curve with equation $y = x^2 - \frac{16}{x}$

- (b) Find the value of k

$$k = \dots\dots\dots (3)$$

(Total for Question 22 is 7 marks)

PAST PAPER SOLUTION

Q#: 22

Marks: 7

TOPIC: **Differentiation**

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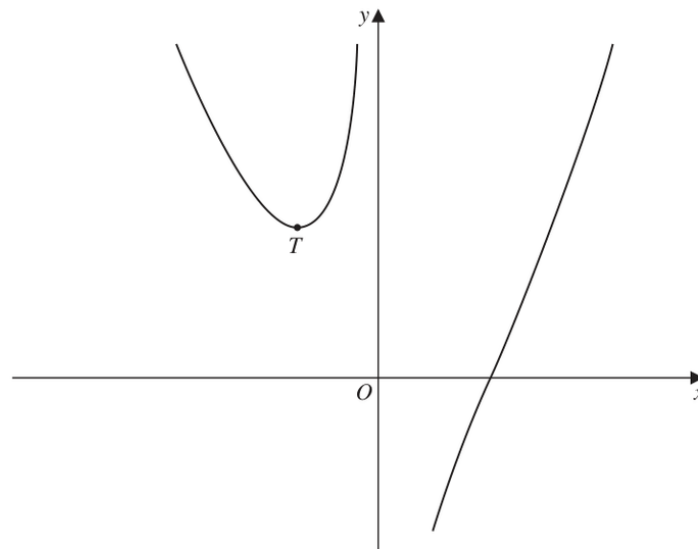
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EA

PAST PAPER QUESTION**Q#: 22****Marks: 7****TOPIC: Differentiation**

Jun 2022 P1HR - Jun 2022 | P1HR

- 22 The diagram shows a sketch of part of the curve with equation $y = x^2 - \frac{p}{x}$ where p is a positive constant.

Diagram **NOT**
accurately drawn

For all values of p , the curve has exactly one turning point and this turning point is a minimum shown as the point T in the sketch.

For the curve where the x coordinate of T is -3

- (a) find the value of p

$$p = \dots\dots\dots (4)$$

The line with equation $y = k$ is a tangent to the curve with equation $y = x^2 - \frac{16}{x}$

- (b) Find the value of k

$$k = \dots\dots\dots (3)$$

(Total for Question 22 is 7 marks)

PAST PAPER SOLUTION

Q#: 22

Marks: 7

TOPIC: **Differentiation**

Jun 2022 P1HR - Jun 2022 | P1HR

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 23****Marks: 5****TOPIC: Completing the Square**

Jun 2022 P1HR - Jun 2022 | P1HR

23 (a) Express $2x^2 - 12x + 3$ in the form $a(x + b)^2 + c$ where a , b and c are integers.

.....
(3)

The curve **C** has equation $y = 2(x + 4)^2 - 12(x + 4) + 3$

The point **M** is the minimum point on **C**

(b) Find the coordinates of **M**

(..... ,)
(2)

(Total for Question 23 is 5 marks)

EA

PAST PAPER SOLUTION

Q#: 23

Marks: 5

TOPIC: **Completing the Square**

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 23****Marks: 5****TOPIC: Completing the Square**

Jun 2022 P1HR - Jun 2022 | P1HR

23 (a) Express $2x^2 - 12x + 3$ in the form $a(x + b)^2 + c$ where a , b and c are integers.

.....
(3)

The curve **C** has equation $y = 2(x + 4)^2 - 12(x + 4) + 3$

The point **M** is the minimum point on **C**

(b) Find the coordinates of **M**

(..... ,)
(2)

(Total for Question 23 is 5 marks)

EA

PAST PAPER SOLUTION

Q#: 23

Marks: 5

TOPIC: **Completing the Square**

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PAST PAPER QUESTION**Q#: 24****Marks: 5****TOPIC: Probability Toolkit**

Jun 2022 P1HR - Jun 2022 | P1HR

24 Elliot has x counters.

Each counter has one red face and one green face.

Elliot spreads all the counters out on a table and sees that the number of counters showing a red face is 5

Elliot then picks at random one of the counters and turns the counter over.
He then picks at random a second counter and turns the counter over.The probability that there are still 5 counters showing a red face is $\frac{19}{32}$ Work out the value of x
Show clear algebraic working. $x = \dots\dots\dots$

(Total for Question 24 is 5 marks)

E.A.

PAST PAPER SOLUTION

Q#: 24

Marks: 5

TOPIC: **Probability Toolkit**

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 24****Marks: 5****TOPIC: Probability Toolkit**

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Each counter has one red face and one green face.

Elliot spreads all the counters out on a table and sees that the number of counters showing a red face is 5

Elliot then picks at random one of the counters and turns the counter over.
He then picks at random a second counter and turns the counter over.The probability that there are still 5 counters showing a red face is $\frac{19}{32}$ Work out the value of x
Show clear algebraic working. $x = \dots\dots\dots$

(Total for Question 24 is 5 marks)

E.A.

PAST PAPER SOLUTION

Q#: 24

Marks: 5

TOPIC: **Probability Toolkit**

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 25****Marks: 5****TOPIC: Sequences**

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25 The sum of the first 10 terms of an arithmetic series is 4 times the sum of the first 5 terms of the same series.

The 8th term of this series is 45

Find the first term of this series.
Show clear algebraic working.

(Total for Question 25 is 5 marks)

EA

PAST PAPER SOLUTION

Q#: 25

Marks: 5

TOPIC: **Sequences**

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 25****Marks: 5****TOPIC: Sequences**

Jun 2022 P1HR - Jun 2022 | P1HR

25 The sum of the first 10 terms of an arithmetic series is 4 times the sum of the first 5 terms of the same series.

The 8th term of this series is 45

Find the first term of this series.
Show clear algebraic working.

(Total for Question 25 is 5 marks)

EA

PAST PAPER SOLUTION

Q#: 25

Marks: 5

TOPIC: **Sequences**

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